

Data and AI Model Governance

Complete Tutorial: Data Governance, Model Risk Management, Validation, GenAI Governance

GARP RAI Exam Preparation

Slide 1: Introduction to Model Governance

What is Model Governance?

- Sound governance of models is essential for effective and prudent risk management
- Without it, a range of outcomes and decisions may occur that can prove detrimental
- Model governance should exist across the entire model lifecycle
- From model development through performance monitoring and ultimately decommissioning

Why Governance Matters for AI/ML:

- Ensures technological advances don't overshadow need for fundamental challenges
- Addresses opacity of AI/ML models
- Proper governance of data used to train these models is particularly relevant
- Generative AI poses additional governance challenges

Key Insight: Governance must span [the entire model lifecycle](#) from development to decommissioning!

Focus: Financial sector and quantitative risk models (QRMs) due to established regulatory framework.

What is Data Governance?

Data Governance Institute Definition:

- "A system of decision rights and accountabilities for information-related processes"
- "Executed according to agreed-upon models which describe who can take what actions"
- "With what information, and when, under what circumstances, using what methods"

Broad Scope:

- Encompasses the people, processes, and technologies required to manage and protect data assets
- Covers entire data lifecycle from collection to deletion
- Balances data accessibility with data protection

Key Components:

- Data strategy and vision
- Data quality management
- Data provenance and legal compliance
- Data classification and security
- Metadata management

Key Insight: Data governance is about **decision rights and accountabilities** for data!

Slide 3: Data Strategy - Vision and Goals

Developing a Data Strategy:

Key Requirements:

- Clear and approved data-governance policy
- Strategic vision and operational practices
- Level of authorization for non-publicly available personal information
- Specification of data usage (single or recurring use)
- Documentation of data sources

Alternative Data Considerations:

- Additional controls and testing required
- Privacy concerns
- Data quality and accuracy risks
- Unbounded data streams

Data Governance Committee Responsibilities:

- Data collection, reconciliations, treatment, access, monitoring
- Management of data stewards
- Overall oversight on all aspects of data governance

Key Insight: Data strategy requires **clear policies and authorization** for data use!

4: Data Quality - Accuracy, Consistency, Integrity

Ensuring Data Quality:

Core Requirements:

- Complete, relevant, transparent, and accurate data should be the primary objective
- Model risk management requirements around data quality do not change with AI/ML algorithms

Alternative Data Sources - Additional Scrutiny:

- Verifying accuracy of alternative data sources may be difficult
- Data providers may change field definitions after deployment
- Even well-established data sources are subject to this risk

Example - COVID-19 Pandemic:

- U.S. Bureau of Labor Statistics unemployment data was biased
- Inability to conduct proper surveys
- Anomalies may occur with any data stream
- New data sources introduce additional risks

Data Monitoring:

- Input data monitoring to alert model owners of flawed inputs
- Outlier detection may be insufficient to spot anomalies
- Definitional changes may only be disclosed in footnotes

Key Insight: Data quality is **critical and requires continuous monitoring!**

Slide 5: Data Provenance

Data Provenance - Verifying Legal Right to Use Data:

Definition:

- Provenance refers to the sequence of ownership and handling of items of value
- Now applies to data used to create models
- Proving legal right to use data is essential

Why It Matters:

- Scrutiny of whether vendors have legal right to use data
- Copyright and intellectual property lawsuits against GenAI vendors
- Example: Data scraped from websites without permission

Regulatory Consequences:

- FTC: Any model built on improperly obtained data must be destroyed
- First such legal action has already been settled with model destruction
- No model can be approved without proving ownership of input data
- Vendor models need to provide proof and indemnify licensees

Key Challenge:

- Established data (e.g., FICO) is generally safe
- Alternative data requires scrutiny
- Risk must be considered for GenAI models where model risk management applies

Key Insight: Data provenance is **essential for legal compliance and model approval!**

Slide 6: Data Classification

Data Classification - Structure and Confidentiality:

Data Types:

1 Quantitative Data:

- Measured or determined numerically
- Examples: income, age, temperature

2 Qualitative Data:

- Determined non-numerically
- Examples: gender, ethnicity, job title

Data Sources:

- Observed data
- Projected data
- Judgment-based data

Critical Governance Requirement:

- Distinguish the classification of data being used
- Ensure access and use is permissible
- When data is confidential or personally identifiable, controls must be:
 - Present
 - Operational
 - Effective

Key Insight: Data classification determines [access controls and usage permissions](#)!

Slide 7: Metadata Management

Metadata Management:

What is Metadata?

- Basic information about data
- Includes: file type, time of creation, data author
- Data source, file size, and other relevant information

Types of Metadata:

- 1 **Descriptive:** Identifies and describes data
- 2 **Structural:** How data is organized
- 3 **Administrative:** Technical and rights information
- 4 **Reference:** Standards and definitions
- 5 **Statistical:** Data quality and methodology

Robust Metadata Management Strategy:

- Data documentation
- Data mapping
- Data dictionary
- Data definitions
- Data process flows
- Data relationships to other data
- Data structures

Key Insight: Metadata management enables **better-informed business decisions!**

Slide 8: Data Security and Protection

Data Security and Protection:

Key Concepts:

- Data security often used interchangeably with data protection and data integrity
- Strategic security process and data protection plan are essential
- Procedural steps identified to safeguard:
 - Availability of data
 - Access to data
 - Data privacy

Data Protection Strategy:

- Safeguarding important information from:
 - Corruption
 - Malicious or accidental damage
 - Compromise or loss
- Importance increases with amount of data created and stored
- Data retention policy should be in place and adhered to

US Regulatory Examples:

- Safeguards Rule (FTC): Financial institutions must keep customer information secure
- Gramm-Leach-Bliley Act (GLBA): Privacy practices and opt-out rights
- Privacy Act of 1974: Federal agency data collection and use

Key Insight: Data security requires [comprehensive protection strategies and regulatory compliance!](#)

Data Access: Permissions and Secure Sharing:

Most Important Aspect:

- Ensuring data is readily accessible and shareable
- To those who have requisite permission and need to access it

Components of Data Access:

- Data security
- Data protection
- Data permissions
- Data access controls

Regulatory Requirements:

- Organizations must implement appropriate security measures
- Encryption to safeguard data
- Ensure confidentiality and integrity
- Maintain compliance

Key Insight: Data access balances [availability with security and permissions](#)!

Compliance: Legal and Regulatory Requirements:

Three Basic Principles:

1 Obtaining Consent:

- Data subjects must consent to data collection
- Informed consent is essential
- Consent can be withdrawn

2 Data Minimization:

- Collect only necessary data
- Most effective privacy protection
- Reduce risk and exposure

3 Ensuring Rights of Data Subjects:

- Right to access data
- Right to correct data
- Right to delete data
- Right to be forgotten

Key Regulatory Frameworks:

- GDPR (Europe)
- CCPA (California)
- PIPL (China)
- GLBA (US Financial)

Key Insight: Data compliance centers on [consent](#), [minimization](#), and [data subject rights](#)!

Slide 11: Board Responsibilities in Data Governance

Board of Directors' Role in Data Governance:

Crucial Responsibilities:

1 Approval of Framework:

- Overall data governance framework and policies
- Clear guidelines on data quality, privacy, security
- Compliance with relevant regulations

2 Oversight of Compliance and Risk Management:

- Compliance with legal, regulatory, and ethical standards
- Data protection laws (GDPR), industry standards, internal policies

3 Risk Assessment and Management:

- Data breaches
- Data quality issues
- Misuse of data

4 Regular Review:

- Adapt to changing business needs
- New technologies
- Regulatory requirements

Key Insight: Board sets the "tone at the top" for data governance culture!

Synthetic Data and AI Readiness:

Synthetic Data Usage:

- Increasing use to augment available data
- Mitigates data risks (client or personally identifiable data)
- Can be designed to avoid biases

Advantages of Synthetic Data:

- No privacy risks (doesn't come from real data subjects)
- Can be more easily rid of problematic biases by design
- Scalable and customizable

Disadvantages:

- May not be as precise as real data (especially visual data)
- May differ from real data in ways not obvious
- Requires careful validation

AI-Ready Data:

- Accurate, unbiased, well-structured
- Cleaned and suitable for training
- Improves accuracy and efficiency of AI models

Key Insight: Synthetic data and AI readiness [require consideration in data strategy!](#)

Slide 13: Data Governance Process

The Data Governance Process:

Key Steps:

1 Data Strategy Development:

- Vision and goals
- Policies and procedures
- Authorization levels

2 Data Collection:

- Identify sources
- Verify provenance
- Ensure legal compliance

3 Data Quality Management:

- Accuracy checks
- Consistency validation
- Integrity verification

4 Data Classification:

- Structure and confidentiality
- Access controls

5 Metadata Management:

- Documentation
- Data dictionary
- Data mapping

6 Data Security and Compliance:

- Protection measures
- Regulatory compliance
- Monitoring and auditing

Key Insight: Data governance is a [continuous process across the data lifecycle!](#)

Slide 14: Roles and Responsibilities in Data Governance

Key Roles in Data Governance:

Board of Directors:

- Approve framework and policies
- Oversee compliance and risk management
- Ensure regular review and updates

Data Governance Committee:

- Overall oversight on all aspects
- Approve data practices and policies
- Not responsible for operational aspects

Data Stewards:

- Day-to-day data management
- Ensure data quality and consistency
- Implement policies

Data Owners:

- Accountability for specific data assets
- Define access permissions
- Ensure proper use

Key Insight: Clear roles and responsibilities are **essential for effective data governance!**

Slide 15: Data Governance Summary

Data Governance: Key Takeaways

Core Components:

1 Strategy:

- Vision and goals
- Policies and authorization
- Alternative data controls

2 Quality:

- Accuracy, consistency, integrity
- Continuous monitoring
- Alternative data scrutiny

3 Provenance:

- Legal right to use data
- Copyright compliance
- Vendor verification

4 Classification:

- Structure and confidentiality
- Access controls

5 Security:

- Protection measures
- Regulatory compliance
- Data retention

Key Insight: Data governance is the **foundation for responsible AI!**

Slide 16: What is a Model?

Defining a Model (SR 11-7):

SR 11-7 Definition:

- "The term 'model' refers to a quantitative method, system, or approach"
- "That applies statistical, economic, financial, or mathematical theories"
- "Techniques, and assumptions to process input data into quantitative estimates"

Key Elements:

1 Quantity of Interest:

- Numerical object whose future value is uncertain
- Examples: portfolio value, revenue projection, client count

2 Potential Future Scenarios:

- Possible values of the quantity of interest
- Each scenario assigned a weight (probability)

3 Risk Measure:

- Summarizes essential information
- Examples: VaR, expected shortfall, average outcome

Key Insight: A model is a **quantitative method for processing data into estimates!**

Slide 17: Risk Modeling Elements

Three Main Elements of Quantitative Risk Modeling:

1. Quantity of Interest:

- Numerical object whose future value is uncertain
- Examples:
 - Portfolio value in ten business days
 - Revenue projection over next five years
 - Expected number of new clients by year-end

2. Potential Future Scenarios:

- Possible values of the quantity of interest
- Each scenario assigned a weight (probability)
- Examples:
 - Portfolio value conditioned on investment decisions
 - Revenue after business model changes
 - New clients after advertising campaign

3. Risk Measure:

- Summarizes essential information from scenarios
- Examples:
 - Value at Risk (VaR)
 - Expected Shortfall (ES)
 - Average outcome across all scenarios

Key Insight: Risk models provide **structured approach to envision the future!**

Key Considerations in Risk Modeling:

1. Completeness of Scenario Sets:

- Challenging to anticipate every potential future scenario
- Historical data may not reflect rare events
- Expert judgment can be difficult

2. Feedback Effects:

- Scenarios and decisions may influence other market participants
- Scenario sets and weights may need ongoing updates

3. Communication of Results:

- Reports should provide summary of main assumptions
- Ensure transparency and avoid complacency
- Reflect perceived risk based on uncertainty and exposure

4. Regulatory Context:

- SR 11-7 (2011) established regulatory standards
- OCC Handbook (2021) includes AI/ML perspectives

Key Insight: Effective risk modeling requires [completeness](#), [awareness of feedback](#), and [clear communication](#)!

Model Governance Policies - Key Elements:

Core Policy Areas:

1 Defining What Constitutes a Model:

- Distinction between model and end-user tool
- Gray areas need consistent adherence
- Presence of uncertainty may indicate a model

2 Model Risk Definition:

- Agree on internal definition
- Align with regulatory requirements

3 Risk Tier Identification:

- Drives depth and frequency of validation
- Materiality, complexity, exposure

4 Model Validation Definition:

- Establish expectations for validation process
- Documentation standards

5 Model Risk Appetite:

- Document institution's risk appetite
- Outline procedures for unacceptable risks

Key Insight: Written policies are the **foundation of model governance frameworks!**

Model Documentation - SR 11-7 Guidance:

Key Requirements:

- Strong governance includes documentation of model development and validation
- Sufficiently detailed to allow parties unfamiliar with a model to understand:
 - How the model operates
 - Limitations of the model
 - Key assumptions of the model

Documentation Best Practices:

- 1 **Comprehensive:**
 - Cover all aspects of model lifecycle
 - Include development and validation
 - Document all key decisions
- 2 **Accessible:**
 - Clear language
 - Well-organized
 - Executive summaries
- 3 **Maintained:**
 - Updated with changes
 - Version control
 - Historical records

Key Insight: Documentation must be sufficiently detailed for others to understand the model!

Slide 21: Model Inventory

Model Inventory - Comprehensive Tracking:

What to Include:

1 Model Identification:

- Model owner, user, and developer information
- Purpose and brief description
- Applicable products

2 Methodology:

- Description of model methodology
- Model classification (e.g., Monte Carlo, logistic regression)
- Whether it's an AI/ML model

3 Development:

- In-house or vendor developed
- Use and frequency
- Restrictions and limitations

4 Assumptions and Limitations:

- Key assumptions
- Known weaknesses
- Management overlays

5 Documentation and History:

- Locations of documentation and validation reports
- Access to model code and databases
- History of updates, approvals, validations

Key Insight: Model inventory enhances **transparency and tracking** of all models!

Model Landscape - Understanding Interdependencies:

Why Model Landscape Matters:

- Models should not be viewed in isolation
- Outputs from one model often serve as inputs for others
- Transparency regarding interdependencies is vital

Key Considerations:

1 Interdependencies:

- Upstream and downstream dependencies
- Impact on key business decisions
- Critical models identification

2 Aggregate Risk:

- Influenced by interdependencies
- Shared assumptions and methodologies
- Models that can affect multiple outcomes

3 Duplication:

- Similar models used in different parts
- Different assumptions
- Inefficiency and risk

Key Insight: Model landscape shows **how models interact and affect each other!**

Registering AI/ML Models - New Aspects:

1. Complexity of Methodology and Design:

- AI/ML models learn autonomously in developer-defined range
- Consider: chosen methodologies (neural networks vs. linear regression)
- Interpretability indicators (number of hidden layers, parameters)

2. Data Usage:

- Volume of required data or number of features
- Complexity of data structures (unlabeled, metadata)
- Data quality (poorly labeled, low quality, unstructured)
- Variable interactions and transformations

3. Output Parameters:

- Supervised vs. unsupervised learning
- Prediction vs. sentiment analysis vs. recommendations
- No direct way to evaluate output accuracy for unsupervised

Key Insight: AI/ML registration requires [additional considerations beyond traditional models!](#)

Slide 24: Model Risk Ranking

Model Risk Ranking Factors:

Key Factors:

1 Materiality:

- Economic consequences of misuse
- Operational impact
- Reputational consequences
- Usage by different parties
- Impact on decisions, financial statements, or regulatory reporting

2 Complexity:

- May elevate risk ranking
- Even if materiality is low
- AI/ML models often have high complexity

3 Exposure:

- If model affects published financial reports
- Risk ranking may be elevated
- Even simple models can have high exposure

Important Note:

- Minimum standards of model risk assessment should apply to all models
- Regardless of materiality
- Absence of assessment introduces its own model risk

Key Insight: Risk ranking considers **materiality, complexity, and exposure!**

Slide 25: Model Performance Monitoring Metrics

Performance Monitoring Metrics:

Classification Models:

- Accuracy
- Precision
- Recall
- F1 Score
- Area Under the ROC Curve (AUC)
- Confusion Matrices

Regression Models:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Coefficient of Determination (R-squared)
- Root Mean Square Error (RMSE)

Additional Metrics:

- Clustering metrics
- Deep learning metrics
- GenAI performance metrics (see Chapter 10, Module 2)

Key Point:

- Performance monitoring must be tailored to model type
- Metric must be appropriate for the model
- Model validation should assess this

Key Insight: Performance metrics must be [appropriate for the specific model type!](#)

Three Lines of Defense in Model Risk Management:

First Line (Business and Model Owners/Developers):

- Owns the model and its performance
- Responsible for designing and implementing models
- Operates daily/weekly performance dashboards
- Owns data quality for model inputs
- First to react to threshold breaches

Second Line (Independent Risk Management/Model Validation):

- Sets model risk management policy
- Conducts independent model validations
- Challenges First Line's performance assessments
- Maintains enterprise-wide model inventory

Third Line (Internal Audit):

- Provides independent assurance to the board
- Reviews whether monitoring is appropriately archived
- Ensures policies are consistently followed

Key Insight: Three lines provide **checks and balances in model risk management!**

Slide 27: First Line - Model Owners and Developers

First Line of Defense - Detailed Responsibilities:

Model Owners:

- Accountable for proper development, validation, approval
- Findings remediation, implementation, use
- Ongoing performance monitoring and reporting
- Change management
- Capture model characterization in inventory
- Assess model materiality

Model Developers:

- Central to creation of models
- Decisions throughout modeling cycle affect risk
- Proper documentation saves time
- Act as both producers and recipients of validation feedback

AI/ML Considerations:

- Ensure continuous learning models don't drift into unvalidated states
- Work closely with MLOps teams
- Data scientists must interpret metrics in business context

Key Insight: Model owners and developers are **responsible for model performance and accountability!**

Slide 28: Second Line - Model Risk Management

Second Line of Defense - Model Risk Function:

Key Responsibilities:

1 Framework Development:

- Develop and maintain model risk management framework
- Establish general validation standards

2 Model Inventory:

- Maintain model inventory
- Combined with material aspects of model landscape

3 Materiality Thresholds:

- Set and maintain thresholds for model materiality
- Related to business decisions

4 Validation Activities:

- Ensure risk model validation activities occur
- Report validation results to board

AI/ML Considerations:

- Deep expertise in validating complex ML models
- Assessing novel performance metrics
- Data drift detection for high-dimensional data
- Fairness/bias assessments
- Tools to "look inside the black box"

Key Insight: Second Line provides **independent challenge and oversight** of models!

Third Line of Defense - Internal Audit:

Key Responsibilities:

1 Independent Assurance:

- Provide assurance to the board
- Model risk management framework is well-designed
- Operating effectively

2 Annual Reviews:

- Monitoring evidence appropriately archived
- Performance exceptions escalated according to policy
- Policies consistently followed and up-to-date

AI/ML Considerations:

- Audit teams need sufficient understanding of ML concepts
- Focus on robustness of controls over automated ML pipelines
- Data governance for ML training data
- Adequacy of Second Line challenge for novel ML approaches

Key Point:

- Audit is "meta-validation" or "validation of validation"
- Not the primary model risk management function
- Vital and necessary independent check

Key Insight: Internal Audit provides [independent verification](#) of the entire MRM framework!

Slide 30: Communication and Interaction

Communication and Interaction in Model Governance:

Key Areas:

1 Breach Notification:

- How does First Line formally communicate performance breaches?
- What information must be provided?

2 Independent Review:

- Second Line's responsibilities and timelines
- Verifying issues and assessing materiality
- Challenging root cause analysis and remediation plans

3 Dispute Resolution:

- If First and Second Lines disagree
- Documented escalation path
- Model Risk Committee or Chief Risk Officer

4 Reporting:

- How findings flow to senior management and board?
- Consolidated and understandable manner

Common Failure Points:

- First Line incentivized to downplay poor performance
- Second Line lacking business context or technical depth
- Third Line focusing on process rather than substance

Key Insight: Effective communication is **vital for successful model governance!**

Model Development Process - Overview:

Key Steps:

1 Define Objectives and Scope:

- Clear problem or risk being addressed
- Applicable products
- Required data and desired outcomes

2 Data Collection and Preprocessing:

- Gather relevant data
- Clean and transform data
- Handle missing values or outliers

3 Data Analysis and Visualization:

- Gain insights into data
- Identify patterns or relationships
- Understand limitations or biases

4 Feature Engineering:

- Select and engineer appropriate features
- Transform or combine variables
- Dimensionality reduction

Key Insight: Model development is a **systematic process from objectives to implementation!**

Model Development Process (Continued):

5. Model Selection:

- Choose modeling approach
- Comprehensive evaluation of alternatives
- Document rationale

6. Model Training/Calibration:

- Optimize parameters and hyperparameters
- Best fit data and minimize errors
- Avoid data leakage (using future or concurrent data)
- Use cross-validation

7. Developer Testing:

- Thorough testing before deployment
- Evaluate performance
- Identify issues early

8. Documentation:

- Clear statement of purpose
- Sound design, theory, and logic
- Robust methodologies
- Rigorous data quality assessment

Key Insight: SR 11-7 requires [sound development process and thorough testing](#)!

SR 11-7 Guidance on Model Development:

Key Requirements:

- Clear statement of purpose to ensure model is developed in line with its intended use
- Sound design, theory, and logic underlying the model
- Robust model methodologies and processing components
- Rigorous assessment of data quality and relevance
- Appropriate documentation

Testing Requirements:

- Integral part of model development
- Various components and overall functioning evaluated
- Show model is performing as intended
- Demonstrate accuracy, robustness, and stability
- Evaluate limitations and assumptions

Important Note:

- Organizations should ensure development of judgmental and qualitative aspects is also sound

Key Insight: SR 11-7 requires [comprehensive development and testing of all model aspects!](#)

Slide 34: Software Testing Types

Types of Software Testing:

Testing Chronological Order:

1 Unit Tests:

- Validate each piece of code performs as designed

2 Component Tests:

- Verify functionality of individual code sections

3 Integration Tests:

- Verify interfaces between components

4 System Tests:

- Ensure completely integrated system meets requirements

5 Performance Tests:

- Assess speed, responsiveness, and stability

6 Regression Tests:

- Verify system continues to function after modifications

7 Acceptance Tests:

- Determine if model is ready for deployment

8 Adversarial Testing:

- "Red team" evaluation of AI models
- Extreme scenarios and edge cases
- Prompt injection and unexpected queries

Key Insight: Testing should be **comprehensive and start as early as possible!**

Slide 35: White-Box vs Black-Box Testing

Testing Approaches:

White-Box Testing:

- Testers have access to internal data structures, algorithms, and code
- Testing at unit level from developer's perspective
- Line-by-line proofreading of code
- More effective for finding specific issues

Black-Box Testing:

- Treats software as a closed box
- From user's perspective
- No knowledge of internal implementation
- Reduces likelihood of bias

Gray-Box Testing:

- Between white-box and black-box
- Informed by knowledge about functional specifications
- Knowledge of architecture of underlying code
- Does not access the code itself

When to Use:

- White-box: internally developed models, early development
- Black-box: proprietary vendor models, functional testing
- Gray-box: distributed applications, testing component integration

Key Insight: Testing approach depends on [model type](#), [development stage](#), and [accessibility](#)!

Effective Test Configurations:

Key Principles:

- Tests need to be appropriately configured
- Usefulness hinges on adequacy and sufficiency of configurations
- Testers should not limit configurations solely to artificial cases
- Most valuable test configurations emerge from real-world situations

Less Realistic Parameters (Better Testing):

- Missing trades
- Excessive trades
- Incorrect scales or mappings
- Extreme parameter values
- Parameters estimated from insufficient data
- Data provided by inexperienced users
- Variations in compilers or hardware

Recommendation:

- Establish preliminary process that automatically generates test results from available data

Key Insight: The less realistic the test parameters, **the better the testing!**

The Use Test - Qualitative Assessment:

What is the Use Test?

- Examines model within its context
- Considers human interaction and actual usage
- Assesses acceptance of the model
- Evaluates interpretation and application of results

Key Focus Areas:

1 People, Not Numbers:

- Focuses on credible stories of acceptance
- Success and failure stories
- Technical challenges take a back seat

2 Crucial Questions:

- Are results utilized appropriately?
- Do they prove useful?
- If not, why?

3 Exploration Areas:

- Acceptance and trust
- Comprehensibility of results
- Communication of feedback to modelers

Key Insight: Use test focuses on **how people interact with and use the model!**

Use Test - Reporting and Impact:

Reporting:

- Results typically presented to senior management
- Not documented as technical reports or spreadsheets
- Easier to communicate results
- Minimizes risk of being seen as arduous requirement

Organizational Impact:

- Contributes to improving culture of risk modeling
- Enhances risk management culture
- Bridges gap between technical and business perspectives

Benefits:

- Promotes acceptance and trust
- Identifies usability issues
- Improves model communication
- Drives continuous improvement

Key Point:

- Use test is qualitative in nature
- Difficult to follow a standard procedure
- Requires ongoing commitment

Key Insight: Use test reporting focuses on [senior management communication and culture improvement!](#)

Should There Be Limits on Model Use?

Why Limits Are Needed:

- Every QRM is based on underlying assumptions
- Assumptions should be clearly stated
- Identify model limitations and weaknesses
- Prevent misuse

Types of Limits:

1 Product Applicability:

- Model approved for specific products only
- Example: Standard interest rate swaptions only
- Not for Bermudan swaption without additional validation

2 Access Restrictions:

- Who can use the model
- Training requirements
- Authorization levels

3 Portfolio Size Limits:

- Particularly for new markets or instruments
- Limited data or experience
- Risk ranking can change as portfolio grows

Key Insight: Model limits prevent misuse and ensure risk ranking stays appropriate!

Three General Guidelines for Risk Modeling Culture:

1. Awareness:

- Be aware of limitations and assumptions of risk modeling
- Understand your company's history with QRMs
- Know risks entailed and validation processes in place
- Stay informed about market practices
- Recognize that the world is constantly changing

2. Transparency:

- Transparently communicate assumptions, limitations, and documentation
- Provide detailed documentation with executive summaries
- Document decision-making process
- Document all validation activities, including unsuccessful attempts
- Engage in open communication with end users

3. Experience:

- Learn from past modeling endeavors
- Apply relevant lessons
- Emphasize proper project management
- Develop prototypes early
- Establish and maintain libraries of reusable code
- Seek input from other modelers and external experts

Key Insight: Guidelines foster **culture that views validation as opportunity for insight!**

Adapting SR 11-7 Guidelines:

Key Principle:

- SR 11-7 established regulatory guidance
- Each institution needs to adapt to specific situation

Factors to Consider:

1 Specific Models:

- Type of models used
- Complexity and risk profile

2 Institution Size and Structure:

- Large vs. small institution
- Organizational structure
- Resources available

3 Governance Framework:

- Existing governance structures
- Board and committee oversight

4 Regulatory Requirements:

- Specific regulatory expectations
- Jurisdictional requirements

5 Budget Constraints:

- Resources allocated to model risk
- Cost-benefit considerations

Key Insight: SR 11-7 is general guidance that **must be adapted to each institution's situation!**

Slide 42: Model Changes Process

Model Changes - Governance Process:

1. Change Management Process:

- Structured process for making changes
- Steps: change requests, impact assessment, documentation, validation, approval

2. Documentation:

- Comprehensive documentation required
- Reason for change, modifications, validation results, approvals

3. Independent Review:

- Ongoing validation of changes
- Performed by individuals not involved in modification
- Assess accuracy, reliability, compliance, and risks

4. Approval:

- Final approval through defined governance process
- Detailed documentation in model inventory
- Rationale, decision, conditions, or limitations

5. Review of Proposed Changes:

- Model owner communicates change to MRM team
- Assessment for materiality
- Change-based validation if material

Key Insight: Model changes require [structured governance and independent review!](#)

AI/ML Review Framework Differences:

1. Frequency:

- More frequent monitoring for AI/ML models
- Higher complexity with more parameters
- Undesirable outcomes under certain conditions
- Greater susceptibility to data drift
- Monitoring dashboards may be useful

2. Validation Stakeholders:

- Wider range of stakeholders affected
- Data protection regulation and ethical policies
- Include HR, compliance, operational risk
- Approval structure modifications needed
- Higher familiarity with AI/ML necessary

3. Validation Content:

- Focus on explainability
- Benchmarking against traditional models
- Assessment based on different data sets
- Assessment of specific cases (edge cases)
- Back testing and cross-validation
- Reporting component for bias detection

Key Insight: AI/ML models require [more frequent review and broader stakeholder involvement!](#)

Implementation and Monitoring of Model Changes:

Implementation:

- Approved changes implemented in controlled manner
- Proper documentation
- Include changes to performance monitoring

Monitoring:

- Ongoing monitoring continues after implementation
- Ensure changes achieve intended objectives
- Remain compliant with regulatory requirements

AI/ML Automation:

- Growing volume of AI/ML models
- Potential need for more frequent retraining
- Greater impact of model errors
- Automated ongoing controls become more relevant

Key Point:

- Automated controls where practical and responsible
- Balance automation with oversight

Key Insight: Model changes must be **implemented with controls and continuously monitored!**

Slide 45: Model Changes Summary

Model Changes - Key Takeaways:

Key Principles:

1 Structured Process:

- Change requests, assessment, validation, approval

2 Documentation:

- Comprehensive record of all changes
- Rationale and approvals

3 Independent Review:

- Ongoing validation
- Objective assessment

4 Frequency:

- AI/ML models need more frequent review
- Address data drift and complexity

5 Stakeholders:

- Broader range of stakeholders
- Include compliance, HR, operational risk

6 Monitoring:

- Continuous monitoring after changes
- Automated controls where practical

Key Insight: Model changes require [comprehensive governance throughout the lifecycle](#)!

What is Model Validation?

Definition:

- Process of rigorously testing the model, inputs, outputs, governance, etc.
- To try to identify the extent of residual model risk
- Assess mitigating controls

Key Aspects:

- Critical component in identifying model risk
- Involves qualitative and quantitative aspects
- Follows model lifecycle

When Validation Occurs:

1 Initial Validation:

- Prior to implementation and usage
- Baseline assessment

2 Routine Periodic Validation:

- Annual reviews
- In-depth periodic baseline revalidations

3 Change-Based Validation:

- Triggered by material changes
- When model owner makes modifications

Key Insight: Model validation is the rigorous testing to identify residual risk!

Model Validation Lifecycle:

Key Stages:

- 1 **Model Identification:**
 - Recognize and classify the model
- 2 **Inventory:**
 - Record in model inventory
 - Schedule for validation
- 3 **Initial Validation:**
 - Baseline validation prior to implementation
- 4 **Production and Monitoring:**
 - Back testing and performance monitoring
 - Ongoing throughout production usage
- 5 **Periodic Validation:**
 - Routine validations
 - Annual or more frequent
- 6 **Change-Based Validation:**
 - Triggered by material changes
- 7 **Retirement:**
 - Store in retired model inventory
 - Decommission

Key Insight: Validation follows the [entire model lifecycle from identification to retirement!](#)

Model Validation Activities Include:

Key Review Areas:

1 Data Assessment:

- Appropriateness of data sources
- Data analysis (proxies, time series lengths)

2 Assumptions and Parameters:

- Assessment of any model assumptions
- Parameter evaluation

3 Limitations:

- Identification of limitations, strengths, weaknesses
- Approved use cases and users

4 Implementation:

- Appropriateness of implementation platform
- Usage of expert judgment and overlays

5 Conceptual Soundness:

- Review of conceptual soundness
- Assessment of benchmarks and best practices

6 Performance Monitoring:

- Analysis of ongoing monitoring plans
- Frequency, key risk indicators, escalation plans

Key Insight: Validation covers **all aspects of the model and its use!**

Slide 49: Validation Activities (Continued)

Model Validation Activities (Continued):

Additional Review Areas:

1 Documentation:

- Review quality of documentation
- Ensure backup plans exist (internal and external)

2 Controls:

- Review controls around implementation and usage
- Access controls

3 Outcomes Analysis:

- Appropriateness of dependent variable
- Support for business objectives

4 Testing:

- Stress and scenario tests
- Model boundary testing
- Parameter stability

5 Backtesting:

- Appropriate cadence with model usage
- VaR models: daily monitoring
- Credit risk models: monthly
- Asset-liability models: quarterly

Key Insight: Validation includes ongoing testing and monitoring appropriate to model type!

Slide 50: Initial Validation

Initial Validation - Three Key Goals:

Goal 1: Operational Feasibility:

- Checked and confirmed
- Model can run as intended without technical malfunctions
- Preproduction phase to run on intended schedule

Goal 2: Proper Documentation:

- Adheres to firm-wide standards
- Model development standards
- Documentation standards
- Implementation standards
- Model monitoring standards
- Third-party governance standards
- Includes executive summaries

Goal 3: User Training:

- Model users receive sufficient training
- Interpret and utilize results effectively

Key Point:

- Results form basis for official approval (sign-off)
- Assumptions must be justified
- Limitations and weaknesses documented

Key Insight: Initial validation ensures **operational feasibility, documentation, and user training!**

Slide 51: Ongoing Validation

Ongoing (Periodic) Validation:

Primary Objective:

- Observe whether model remains aligned with intended purpose
- Assumptions remain valid
- Data still appropriate
- Performance monitoring indicates expected performance

Key Activities:

1 Methodology Reassessment:

- Still in line with best practices
- Reflects real world

2 Assumption Validation:

- Original assumptions remain valid
- Real world consistently challenges assumptions

3 Benchmark/Challenger Models:

- Support validation
- Compare results

Important Considerations:

- Well documented
- Consolidated within single document
- Regular validation with fixed agenda
- Depth and frequency commensurate with risk tier

Key Insight: Ongoing validation ensures **models remain fit for purpose over time!**

General Issues in Model Validation:

1. Terminology:

- Alternative terms: model review, model QA, model evaluation, model vetting

2. Focus on Suitability:

- Ultimate goal: assess if model is appropriate and effectively used
- Initial validation critical for credibility
- Ongoing performance monitoring as proposed and approved
- Validation must remain within intended application domain

3. Recursive Validation:

- Validation activities should be subject to critical examination
- Ensure validation doesn't rely on defective data
- Must remain within intended application domain
- Need for challenge of validation activities

4. No "Valid" Model:

- Models are simplified representations under assumed conditions
- Goal is not to test for a "valid" model
- Subject model to attempts to invalidate it
- Successful validation = withstood rigorous testing

Key Insight: Validation is about **testing**, not **proving validity**!

Slide 53: Usefulness vs Validity

Usefulness vs. Validity:

Key Principle:

- Models do not have to be perfect to be useful
- If weaknesses and limitations are identified:
 - Insights can improve the model
 - Restrict application
 - Enhance understanding of capabilities and limitations

Important Considerations:

- Model weaknesses can arise from regime shifts
- Need for ongoing monitoring and validation over time

Effective Challenge (SR 11-7):

- Critical analysis by objective and informed parties
- Identify crucial model assumptions and limitations
- Spot and communicate relevant model weaknesses

Requirements for Effective Challenge:

- Independence from model development
- High level of competence
- Sufficient degree of influence

Key Insight: Models don't need to be perfect to be **useful**, but **limitations must be understood!**

Validation of GenAI Models - Special Considerations:

Key Challenges:

1 Open-Ended Output:

- Unstructured input and output
- Makes validation challenging

2 Performance Metrics:

- Different and more complex than traditional AI/ML
- New metrics needed

3 Data Drift and Coverage:

- Responses may become obsolete
- Changing environment
- Model obsolescence

4 Hallucination:

- Tendency to generate factually incorrect responses
- Appear very convincing

5 Discrimination and Bias:

- Ensure responses are not discriminatory or biased
- AI can amplify bias in data

Key Insight: GenAI validation requires [addressing unique challenges of generative models!](#)

Slide 55: GenAI Validation Challenges (Continued)

GenAI Validation Challenges (Continued):

6. Multi-Modal Issues:

- Processing images, video, and audio data
- In addition to text
- More challenging to validate

7. Interpretability and Explainability:

- Results difficult to explain or interpret
- Black-box nature
- Regulatory compliance challenges

8. Rapid Technological Change:

- More frequent review required
- Regular recalibration needed
- Keeping pace with developments

9. Third-Party and Open-Source Models:

- Increasingly offered as models as a service
- Particular attention needed
- Limited transparency

10. Computational Resources:

- Significant resources required
- Training, validation, recalibration
- Infrastructure challenges

Key Insight: Human review is **essential** in GenAI validation!

Slide 56: Model Design Motivation

Model Design - Motivation for Development:

Changing Conditions:

- New markets
- New products
- Customer requests
- Models must be validated

Regulatory Pressure:

- Regulatory findings
- Model not performing as intended
- Redevelopment or replacement needed

Recent Crisis:

- Losses highlight urgency
- Enhanced or more sensitive modeling needed
- Lessons from crisis

Internal Concerns:

- Senior management discomfort
- Changes in business model
- Increased exposure to innovative products
- Changing market conditions

Innovation and Growth:

- New or evolving customer demands
- New products or model extensions
- Business acquisitions

Key Insight: Model development is driven by business needs, regulation, and changing conditions!

Slide 57: Model Developers

Model Developers - The "Quants":

Required Expertise:

- Mathematics
- Econometrics
- Statistics
- Strong product knowledge

Background:

- Advanced academic degrees
- Mathematics, physics, engineering, computer science
- Scientific background

Key Qualities:

- Creative problem-solving
- Skepticism
- Out-of-the-box thinking
- Experience including past crises

Important Balance:

- Most challenging aspect: designing and constructing potential future scenarios
- Assigning weights to scenarios
- Must be mindful of difficulties in risk definition
- Avoid uncritical use of distributions
- Critical reflection on assumptions

Key Insight: Model developers need **quantitative expertise and practical experience!**

Modeling Issues - Discretization:

What is Discretization?

- Describing the world using continuous variables
- Cases where variables are discretized to finite values
- Reducing complexity in modeling path-dependent events
- Events occurring at discrete time intervals

Why Discretize?

- Complex equations without closed-form solutions
- Numerical approximation via discretization
- Should yield very close results when properly implemented
- Time intervals divided into days, months, years

Challenges:

- Discretization error
- Stability
- Convergence
- Computational complexity
- Boundary conditions
- Numerical precision
- Choice of time step is crucial

Key Insight: Discretization enables implementation but [requires careful parameter selection!](#)

Slide 59: Modeling Issues - Approximations

Modeling Issues - Approximations:

What are Approximations?

- Replacing complex or hard-to-evaluate components
- Alternative methods that produce similar results
- More conveniently implemented

Key Considerations:

- Choosing an approximation fixes the precision
- No room for improvement except by selecting different approximation

Examples:

- Price sensitivities
- Taylor series approximation in market risk
- Valid for simple instruments and small changes

Time Horizon Scaling:

- Scaling VaR by square root of time
- Assumes normal distribution and temporal independence
- Often convenient but rarely empirically justified
- Validity must be examined via statistical tests

Key Insight: Approximations simplify but **fix precision and require careful validation!**

Slide 60: Modeling Issues - Numerical Evaluation

Modeling Issues - Numerical Evaluation:

What is Numerical Evaluation?

- Evaluation of model component with potential for arbitrary precision
- Equipped with parameters controlling precision

Methods:

- Computation of integrals
- Evaluating non-elementary functions
- Solving systems of equations
- Linear algebra problems
- Interpolation
- Numerical differential equation solutions
- Optimization

Key Factors:

- **Stability:** Small errors should not lead to unbounded imprecision
- **Convergence:** Results approach true value as precision increases
- **Floating-Point Arithmetic:** Rounding errors may affect stability

Trade-off:

- Running time vs. precision
- Theoretical confirmation rarely sufficient
- Extensive testing necessary

Key Insight: Numerical evaluation requires **balancing precision with computational cost!**

Slide 61: Modeling Issues - Random Numbers

Modeling Issues - Random Numbers:

Why Random Numbers Needed:

- Models rely heavily on probability theory
- Assigning values to random variables
- Samples drawn from distributions

Challenges:

- Generating random numbers is difficult for both humans and computers
- Computers use pseudo-random number generators (RNGs)
- Depend on initial seed
- Cannot produce actual random numbers

Best Practice:

- Rely on well-designed deterministic pseudo-RNGs
- Rather than introducing randomness in choice of RNG

RNG Quality Assessment:

- Specialized test suites
- Criteria: speed and reproducibility

Key Insight: Random number generation requires [high-quality deterministic pseudo-RNGs](#)!

Model Implementation Tasks:

1. Model Design:

- Designed and documented by model designers
- Facilitate translation into computer code

2. Core Implementation:

- Inner workings implemented as black box
- With specified interfaces

3. System Implementation:

- Model core integrated into system
- Provides user interfaces
- Collects input data and parameters
- Schedules computations
- Feeds the core
- Processes output data
- Keeps history of computations

Key Considerations:

- Different skills for core vs system implementation
- Different programming languages
- Specialized skills needed for system implementation

Key Insight: Implementation involves [model design](#), [core development](#), and [system integration](#)!

Slide 63: Model Adaptation Tasks

Model Adaptation Tasks:

Why Adaptation Needed:

- No model lasts forever
- Periodic reviews detect weaknesses
- Demand adaptation or revision

Adaptation Types:

1 Model Adaptation:

- Model and documentation adapted (or replaced)
- Existing computer code adapted

2 Core Adaptation:

- Model core adapted (or replaced)
- Interfaces may be adapted

3 System Adaptation:

- System around model core adapted (or replaced)

Challenges:

- Initial implementation team may have left
- External consultants may have done some tasks
- If model no longer understood, adaptation is challenging
- If same people, risk of ignorance or entrenched poor practices

Key Insight: Adaptation requires [understanding the model and careful change management!](#)

Slide 64: Running a Model

Running a Model - Key Considerations:

Robustness:

- Models can become complex
- Intertwining with multiple systems
- Increased risk of failure
- Minimize complexity
- Eliminate exotic systems
- Standardize interfaces
- Avoid "reinventing the wheel"

Computational Considerations:

- Total response time varies greatly
- Credit risk models: may take weeks
- Market risk models: daily reporting
- Pre-deal checking: faster processing
- Overall running time determined by slowest task
- Over-optimization for specific hardware can be risky

Flexibility:

- Scheduled runs: regular numbers with minimal manual intervention
- Ad hoc runs: additional information on short notice
- Staff knowledgeable about model interfaces
- Prevent gradual shift from scheduled to ad hoc mode

Key Insight: Running models requires **robustness, appropriate computing, and flexibility!**

Slide 65: Model Adaptation and Running Summary

Implementation, Adaptation, and Running - Key Takeaways:

Implementation:

- Model design, core implementation, system integration
- Different skills for different tasks
- Documentation is essential

Adaptation:

- Periodic reviews detect weaknesses
- Model, core, or system adaptation
- Challenges with changing teams

Running:

- Robustness and standardization
- Computational considerations
- Scheduled vs. ad hoc runs

Common Issues:

- Over-optimism
- Inadequate budgeting
- Midstream redefinition of goals
- Misunderstandings and miscommunication
- Programming errors
- Lack of software quality assurance
- Inappropriate design architecture

Key Insight: Model operations require careful management of implementation, adaptation, and running!

Potential Sources of Misinterpretation:

1. Lack of Comprehension:

- Users lack comprehensive understanding of model
- Underlying assumptions unknown
- Limitations of outputs unclear
- Necessary knowledge and expertise missing

2. Neglecting Model Limitations:

- Models have inherent limitations
- May assume certain conditions
- May not fully capture all relevant risks
- Failing to acknowledge limitations leads to flawed understanding

3. Overemphasis on Point Estimates:

- Models provide estimates with uncertainty
- Focusing solely on point estimates
- Ignoring range of potential outcomes
- Understanding probability distributions is crucial

Key Insight: Misinterpretation can have significant implications for decision-making!

Potential Sources of Misinterpretation (Continued):

4. Failure to Consider Qualitative Factors:

- Models focus on quantitative measures
- Qualitative factors should not be overlooked
- Market conditions, regulatory changes, external influences
- Exclusion can lead to distorted understanding

5. Ignoring Decision Context:

- Model results considered independently
- Without specific context of decision
- Models should inform decision making within context
- Incorporate broader business strategies, risk appetite

6. Confirmation Bias:

- Users interpret results to align with preconceived beliefs
- Cloud judgment
- Compromise objective interpretation
- Approach results with open mind and commitment to unbiased analysis

Key Insight: Misinterpretation arises from [cognitive biases and lack of context!](#)

Slide 68: Fostering Proper Interpretation

Fostering Proper Interpretation of Model Results:

Key Practices:

1 Training and Education:

- Ensure users understand model
- Explain assumptions and limitations
- Provide context for results

2 Clear Communication:

- Present results with uncertainty ranges
- Include confidence intervals
- Explain probabilistic interpretations

3 Open Dialogue:

- Encourage questioning of results
- Discuss alternative interpretations
- Foster critical thinking

4 Culture:

- Foster culture of open dialogue
- Encourage critical thinking
- Cross-functional collaboration

Key Point:

- Guard against misinterpretation biases
- Approach results with open mind
- Commit to unbiased analysis

Key Insight: Proper interpretation requires [training](#), [clear communication](#), and [supportive culture](#)!

Recommendations for AI/ML Model Governance:

1. Begin with Existing Frameworks:

- Don't start from scratch
- Use experience from traditional model validation
- Build on a strong foundation

2. Consider the New Role of Data:

- AI/ML is “model-free” – everything depends on data
- Address issues such as:
 - Various forms of bias
 - Overfitting
 - Population drift
 - Model drift and regime changes
 - Risk of harmful output
- New uses will reveal further challenges:
 - Data limitations
 - Biases
 - Privacy and right to use
 - Legality

Key Insight: AI/ML governance must [address the central role of data!](#)

Recommendations for AI/ML Model Governance (Continued):

3. Add New Perspectives:

- AI/ML increases relevance of ethical aspects
- Data bias
- Explainability of model results
- Recalibration process
- These attributes need to be considered in model inventory

4. BIS Survey Findings - Data Governance:

- Adequate documentation
- Policies, standards, procedures
- Data quality - accuracy, transparency, provenance
- Security and responsible use
- Comprehensive data management systems
- Metadata on sources, version, curation
- Asset inventories and metadata registries

Key Insight: AI/ML governance must incorporate ethical considerations and robust data governance!

BIS Survey - Organizational Structures for AI Governance:

Common Approaches:

1 Centralized Solutions:

- Steering committees/commissions
- Program management offices
- Multi-modal structures

2 Hub-and-Spoke Systems:

- Overall network with multiple functional hubs
- Connected to various departments

3 Fully Decentralized:

- Governance at business line level
- Local control

Accountability Assignments:

- Existing roles: Chief Data/Information/Technology Officer
- Existing units: IT department
- Newly created functions
- Informal setups: communities of practice, cross-functional networks

Key Insight: AI governance structures range from **centralized to decentralized with various accountability models!**

BIS Survey - Guidelines and Risk Management:

Guidelines and Policies:

- Terms of use for AI systems
- Modalities, interfaces, and data
- Models, third-party dependencies
- Upstream/downstream data
- Guidelines for responsible AI use
- Privacy standards
- Explainability requirements
- International principles (e.g., Fundamental Principles of Official Statistics)
- Documentation of methods for regular assessment

Risk Management Frameworks:

- Regular audit of AI systems for vulnerabilities
- Evaluation of robustness of safety measures
- Incident response procedures
- Business continuity plans
- Processes to promote and maintain skills
- Prevent erosion of knowledge due to overreliance
- Leverage existing frameworks (three lines of defense)

Key Insight: Comprehensive guidelines and risk management are **essential for AI governance!**

Why GenAI Governance Is Harder:

1. Opacity and Unpredictability:

- Functioning almost always opaque
- Output unpredictable
- Difficult to interpret or audit
- Probabilistic nature: same prompt may lead to different outputs
- Reproducibility challenges

2. Emergent Capabilities:

- Behaviors not anticipated by developers
- As models become larger and more sophisticated
- Skills not observed in smaller models
- Changing capabilities hard to assess

3. Post-Deployment Shifts:

- Model behavior can shift after deployment
- External providers may update or fine-tune models
- Changes in prompts or connected data alter performance
- One-off validation is inadequate

Key Insight: GenAI governance must address [opacity](#), [unpredictability](#), and [evolving capabilities](#)!

Internal Governance Levers for GenAI:

1. Assign Ownership and Accountability:

- Clarify responsibility at each stage
- Tool selection, testing, deployment, monitoring
- Ensure responsibility doesn't fall between teams

2. Risk-Based Review and Deployment:

- Tiered review gates
- Full risk reviews for high-impact systems
- Light checks for low-stakes internal tools

3. Incentivize Responsible Behavior:

- Risk-tied performance metrics
- Reward flagging risks, documenting limitations
- Align team incentives with responsible use

4. Staff Training:

- Equip staff with knowledge of GenAI risks
- Data protection, hallucinations, complaints

Key Insight: Internal governance requires [clear ownership](#), [risk-based reviews](#), and [proper incentives](#)!

External Governance and Adaptive Tools:

External Governance Constraints:

- Track evolving legal and regulatory requirements
- EU AI Act, Digital Services Act, US state laws
- Evaluate responsibilities with model providers
- Understand incident policies and update schedules
- Align with emerging standards (NIST AI RMF, ISO/IEC 42001)

Adaptive Tools and Practices:

- Continuously monitor post-deployment usage
- Use red-teaming and adversarial testing
- Develop scenario plans for unpredictability
- Deploy technical safeguards (watermarking, usage logging)
- Enable feedback loops and update mechanisms

Key Insight: GenAI governance requires [external awareness and adaptive practices](#)!

Strategic Risk Assessment Questions:

Key Questions:

- ❶ Is GenAI the right tool for this application?
 - Ensure reasons for deploying are good reasons
 - Know what trade-offs are to be expected
- ❷ What are the risks to:
 - Safety
 - Legality
 - Reputation
 - User trust
 - Transparency
- ❸ Risk Identification:
 - Identify relevant risks before deployment
 - Map potential unpredictable behaviors
 - Assess possible impacts

Key Point:

- Strategic assessment should occur before deployment
- Consider both intended and unintended consequences

Key Insight: Strategic assessment ensures **GenAI is the right tool for the right application!**

Model Selection and Deployment Questions:

Key Questions:

- 1 What model is being used?
 - What do we know about its training data?
 - What do we know about its alignment?
- 2 Do we control the model?
 - API vs open source
 - Control over updates
 - Transparency varies

Why These Questions Matter:

- Understanding training data reveals potential biases
- Control determines ability to audit and modify
- Transparency affects governance capabilities

Key Point:

- Model selection affects all subsequent governance
- Choose models that align with governance capabilities

Key Insight: Model selection determines **governance options and risk exposure!**

Data Governance Questions:

Key Questions:

- ❶ What data is being used?
 - Is any of it proprietary or sensitive?
 - Legal restrictions (GDPR, etc.)
 - Client confidentiality
 - Intellectual property
- ❷ Are data protection and retention policies in place?
 - Trace how inputs are used by providers
 - Understand data flow

Why These Questions Matter:

- Data provenance affects legal compliance
- Data protection policies prevent breaches
- Understanding data flow enables governance

Key Point:

- Data governance is foundational for GenAI
- Know what data is used and how it's protected

Key Insight: Data governance questions are **essential** for responsible GenAI use!

Technical Risk Control Questions:

Key Questions:

- ❶ Have we tested for unpredictability?
 - Hallucinations
 - Model drift
 - Unexpected behaviors
- ❷ What safeguards are in place against misuse?
 - Guardrails
 - Human review triggers
 - Usage limits
 - Logs and monitoring

Why These Questions Matter:

- Unpredictability can lead to harmful outcomes
- Safeguards prevent misuse and limit damage
- Monitoring enables detection and response

Key Point:

- Technical controls are essential for safe deployment
- Test for edge cases and unexpected behaviors

Key Insight: Technical risk controls are **essential for safe GenAI deployment!**

Organizational Structure and Incentive Questions:

Key Questions:

- ❶ Who owns risk for GenAI?
 - Is that person/team empowered?
 - Clear accountability
- ❷ Are staff appropriately trained?
 - Know of unpredictable behavior
 - Understand risks and limitations
 - Prevent unintended consequences
- ❸ Are incentives aligned with responsible AI use?
 - Not just speed to deploy
 - Risk-aware behaviors rewarded

Why These Questions Matter:

- Ownership ensures accountability
- Training prevents misuse
- Incentives drive behavior

Key Insight: Organizational structure must support responsible GenAI use!

External Environment Questions:

Key Questions:

- 1 Are we prepared for changing laws and regulation?
 - EU AI Act
 - US state legislation
 - Compliance frameworks
- 2 Are we aligned with best practices?
 - NIST AI RMF
 - ISO/IEC 42001
 - Industry standards

Why These Questions Matter:

- Regulatory landscape is evolving
- Compliance is mandatory
- Best practices build trust

Key Point:

- Following recognized frameworks strengthens compliance
- Builds trust with external stakeholders
- Demonstrates commitment to responsible AI

Key Insight: External alignment is **essential** for regulatory compliance and stakeholder trust!

Slide 82: Questions for Risk Professionals Summary

Questions for Risk Professionals - Summary:

Strategic Risk:

- Is GenAI the right tool?
- What are the risks?

Model Selection:

- What model is being used?
- Do we control it?

Data Governance:

- What data is being used?
- Are protection policies in place?

Technical Controls:

- Have we tested for unpredictability?
- What safeguards are in place?

Organizational:

- Who owns risk?
- Are staff trained?
- Are incentives aligned?

External:

- Are we prepared for regulation?
- Are we aligned with best practices?

Key Insight: A structured questioning approach [supports comprehensive GenAI governance!](#)

Slide 83: Final Thoughts - Team Qualification

Final Thoughts - Team Qualification:

Validation Team Requirements:

- Knowledge of AI/ML techniques
- Hands-on expertise
- Strong foundations in:
 - Product knowledge
 - Econometrics
 - Statistics
 - Computer science
- Proactive approach to staying abreast of latest advancements

Additional Considerations:

- Augment validation staff
- Add AI/ML validation team
- Hands-on knowledge of:
 - AI/ML development platforms
 - Cloud-based environments
 - Low-code and no-code environments
 - CI/CD and MLOps platforms

Key Point:

- Familiarity with AI/ML technologies beyond traditional knowledge
- Specific training may be necessary
- Incorporate specialists
- Use external experts if needed

Key Insight: Validation teams need **updated skills and expertise for AI/ML models!**

Final Thoughts - End-to-End Review:

Review Framework:

- Entire model validation framework needs review
- Regulators consider AI/ML models within scope of model risk management
- SR 11-7, being principle-based, applies to AI/ML models

Implications:

- Challenges in model design
- Implications related to:
 - Data
 - Implementation
 - Monitoring
 - Documentation
 - Use

Key Point:

- AI/ML models fall within existing regulatory frameworks
- Comprehensive review is necessary

Key Insight: End-to-end review is **essential for AI/ML model validation!**

Slide 85: Final Thoughts - Balance and GenAI

Final Thoughts - Balance and GenAI:

Model Complexity Balance:

- More complex is not always better
- Balance model performance with:
 - Interpretability
 - Feedback from supervisors
 - Cost and effort of implementation
 - Maintenance and monitoring
 - In-house expertise
 - Availability of reputable code libraries
 - Academic underpinning
- Validation teams need balanced approach:
 - Not just regulatory-driven
 - Not just performance-oriented
 - Nuanced cost-benefit analysis

GenAI Governance:

- Requires particular focus
- Support development of shared standards
- Transparency around GenAI
- Longer-term credibility depends on:
 - Internal compliance
 - Broader governance ecosystem

Key Insight: Balance and GenAI governance are [essential for sustainable model risk management!](#)

Key Points on Model Development and Testing:

Development Steps:

- 1 Define objectives and scope
- 2 Data collection and preprocessing
- 3 Data analysis and visualization
- 4 Feature engineering
- 5 Model selection
- 6 Model training/calibration
- 7 Developer testing
- 8 Documentation

Testing Types:

- Unit tests
- Component tests
- Integration tests
- System tests
- Performance tests
- Regression tests
- Acceptance tests
- Adversarial testing

Key Insight: Model development and testing are **systematic processes requiring comprehensive approaches!**

Slide 87: Model Validation - Key Points

Key Points on Model Validation:

Validation Lifecycle:

- 1 Model identification
- 2 Inventory
- 3 Initial validation
- 4 Production and monitoring
- 5 Periodic validation
- 6 Change-based validation
- 7 Retirement

Validation Activities:

- Data assessment
- Assumptions and parameters
- Limitations identification
- Implementation review
- Conceptual soundness
- Performance monitoring
- Documentation review
- Controls review
- Outcomes analysis
- Testing and backtesting

Key Insight: Validation is a **comprehensive, ongoing process throughout the model lifecycle!**

Key Points on Misinterpretation:

Common Sources:

- ❶ Lack of comprehension
- ❷ Neglecting model limitations
- ❸ Overemphasis on point estimates
- ❹ Failure to consider qualitative factors
- ❺ Ignoring decision context
- ❻ Confirmation bias

Prevention Strategies:

- Training and education
- Clear communication
- Open dialogue
- Supportive culture

Key Point:

- Misinterpretation can have significant implications
- Requires active management
- Culture of critical thinking is essential

Key Insight: Misinterpretation can be prevented through [education](#), [communication](#), and [culture](#)!

Three Lines of Defense - Key Points:

First Line (Model Owners/Developers):

- Own the model and its performance
- Design and implement models
- Operate performance dashboards
- First to react to breaches

Second Line (Risk Management/Validation):

- Set model risk management policy
- Conduct independent validations
- Challenge First Line assessments
- Maintain model inventory

Third Line (Internal Audit):

- Provide independent assurance
- Review framework effectiveness
- Ensure policy compliance

Key Insight: Three lines provide [checks and balances in model risk management!](#)

Model Changes - Key Points:

Change Management Process:

- 1 Change requests
- 2 Impact assessment
- 3 Documentation
- 4 Validation
- 5 Approval
- 6 Implementation
- 7 Monitoring

AI/ML Considerations:

- More frequent monitoring needed
- Broader stakeholder involvement
- Focus on explainability
- Benchmarking against traditional models
- Assessment on different data sets
- Assessment of specific cases
- Back testing and cross-validation
- Reporting for bias detection

Key Insight: AI/ML model changes require [more frequent review and broader stakeholder involvement!](#)

Data Governance - Key Points:

Core Components:

1 Strategy:

- Vision and goals
- Policies and authorization

2 Quality:

- Accuracy, consistency, integrity
- Continuous monitoring

3 Provenance:

- Legal right to use data
- Copyright compliance

4 Classification:

- Structure and confidentiality
- Access controls

5 Security:

- Protection measures
- Regulatory compliance

Key Insight: Data governance is the foundation for responsible AI and model management!

Model Inventory - Key Points:

What to Include:

- 1 Model identification (owner, user, developer)
- 2 Purpose and description
- 3 Methodology and classification
- 4 Development source (in-house/vendor)
- 5 Assumptions and limitations
- 6 Documentation locations
- 7 History of updates and validations
- 8 Performance monitoring frequency
- 9 Regulatory use indication
- 10 Risk tier

AI/ML Additional Considerations:

- Complexity of methodology
- Data usage and quality
- Output parameters
- Model recalibration needs
- Risk ranking factors
- Performance monitoring metrics

Key Insight: Model inventory provides [comprehensive tracking and transparency](#) for all models!

Model Risk Ranking - Key Points:

Key Factors:

1 Materiality:

- Economic consequences
- Operational impact
- Reputational consequences
- Usage by different parties
- Impact on decisions, financial statements, regulatory reporting

2 Complexity:

- May elevate risk ranking
- Even if materiality is low

3 Exposure:

- Published financial reports
- Wide impact
- Regulatory significance

Important Note:

- Minimum standards for all models regardless of materiality
- Absence of assessment introduces its own risk

Key Insight: Risk ranking determines **validation frequency and intensity!**

Performance Monitoring - Key Points:

Classification Models:

- Accuracy, Precision, Recall, F1, AUC
- Confusion Matrices

Regression Models:

- MAE, MSE, R-squared, RMSE

Additional Metrics:

- Clustering metrics
- Deep learning metrics
- GenAI performance metrics

Key Point:

- Performance monitoring must be tailored to model type
- Metrics must be appropriate for the model
- Model validation should assess metric appropriateness

Key Insight: Performance metrics must be [appropriate for the specific model type!](#)

GenAI Validation - Key Points:

Special Considerations:

- 1 Open-ended output
- 2 Different performance metrics
- 3 Data drift and coverage
- 4 Hallucination
- 5 Discrimination and bias
- 6 Multi-modal issues
- 7 Interpretability and explainability
- 8 Rapid technological change
- 9 Third-party and open-source models
- 10 Computational resources

Essential:

- Human review is essential
- Regular validation and recalibration
- Particular attention to third-party models

Key Insight: GenAI validation requires [addressing unique challenges and human review!](#)

GenAI Governance - Key Points:

Internal Levers:

- 1 Clear ownership and accountability
- 2 Risk-based review and deployment
- 3 Incentivize responsible behavior
- 4 Staff training
- 5 Validation and monitoring

External Constraints:

- 1 Track legal and regulatory requirements
- 2 Evaluate responsibilities with providers
- 3 Align with emerging standards

Adaptive Tools:

- 1 Continuous monitoring
- 2 Red-teaming and adversarial testing
- 3 Scenario planning
- 4 Technical safeguards
- 5 Feedback loops

Key Insight: GenAI governance requires [internal](#), [external](#), and [adaptive approaches](#)!

Slide 97: Questions for Risk Professionals - Summary

Questions for Risk Professionals - Summary:

Strategic Risk:

- Is GenAI the right tool?
- What are the risks?

Model Selection:

- What model is being used?
- Do we control it?

Data Governance:

- What data is being used?
- Are protection policies in place?

Technical Controls:

- Have we tested for unpredictability?
- What safeguards are in place?

Organizational:

- Who owns risk?
- Are staff trained?
- Are incentives aligned?

External:

- Are we prepared for regulation?
- Are we aligned with best practices?

Key Insight: Structured questioning [supports comprehensive GenAI governance!](#)

Implementing AI Governance - BIS Survey Findings:

Data Governance:

- Adequate documentation
- Data quality (accuracy, transparency, provenance)
- Security and responsible use
- Comprehensive data management systems
- Metadata on sources, version, curation
- Asset inventories and metadata registries

Organizational Structures:

- Centralized solutions (steering committees, PMOs)
- Hub-and-spoke systems
- Decentralized approaches
- Accountability assignments

Key Insight: BIS survey shows diverse approaches to AI governance implementation!

Implementing AI Governance - BIS Survey (Continued):

Guidelines and Policies:

- Terms of use for AI systems
- Guidelines for responsible AI use
- Privacy standards
- Explainability requirements
- International principles
- Documentation of assessment methods

Risk Management Frameworks:

- Regular audit of AI systems
- Evaluation of safety measures
- Incident response procedures
- Business continuity plans
- Skills maintenance processes
- Leverage existing frameworks (3LoD)

Key Insight: Comprehensive guidelines and risk management are **essential for AI governance!**

Data and AI Model Governance - Comprehensive Summary:

Key Pillars:

- 1 **Data Governance:**
 - Strategy, quality, provenance, classification, security
- 2 **Model Governance:**
 - Policies, documentation, inventory, risk ranking
- 3 **Model Development:**
 - Objectives, data, features, training, testing
- 4 **Model Validation:**
 - Initial, ongoing, change-based
 - Comprehensive review activities
- 5 **Three Lines of Defense:**
 - Owners/developers, risk management, internal audit
- 6 **GenAI Governance:**
 - Internal levers, external constraints, adaptive tools

Exam Success: Understand [data governance](#), [model risk management](#), [validation](#), and [GenAI governance](#)!